

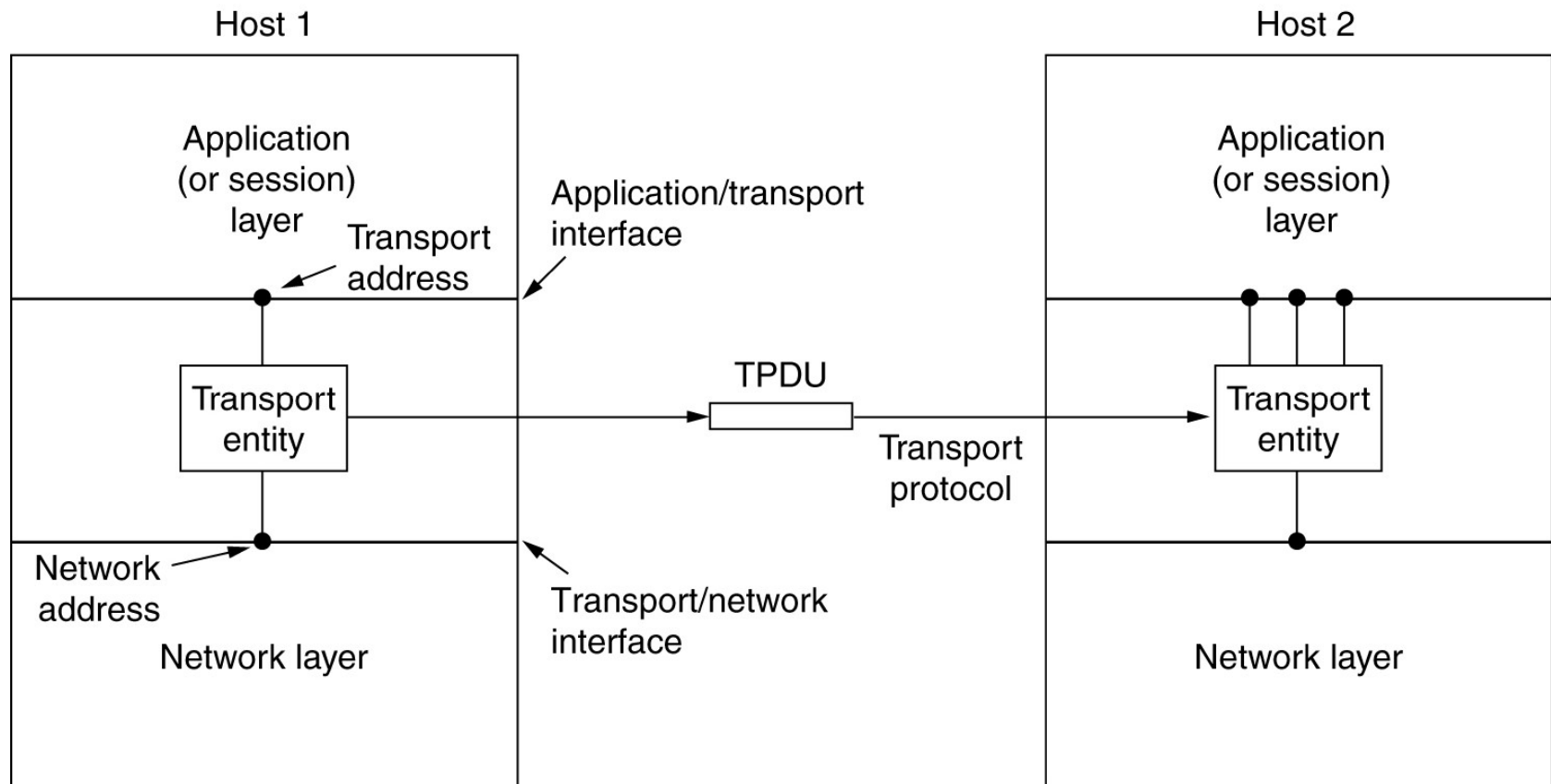
Unit- 6

The Transport Layer

The Transport Service

- Services Provided to the Upper Layers
- Transport Service Primitives
- Berkeley Sockets

Services Provided to the Upper Layers



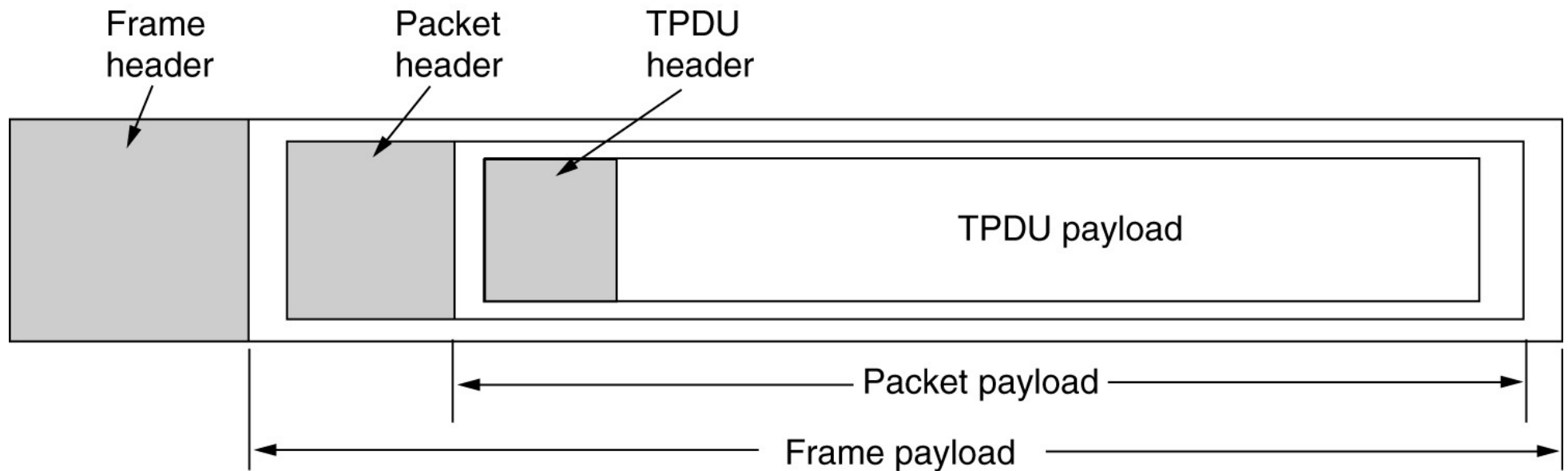
The network, transport, and application layers.

Transport Service Primitives

Primitive	Packet sent	Meaning
LISTEN	(none)	Block until some process tries to connect
CONNECT	CONNECTION REQ.	Actively attempt to establish a connection
SEND	DATA	Send information
RECEIVE	(none)	Block until a DATA packet arrives
DISCONNECT	DISCONNECTION REQ.	This side wants to release the connection

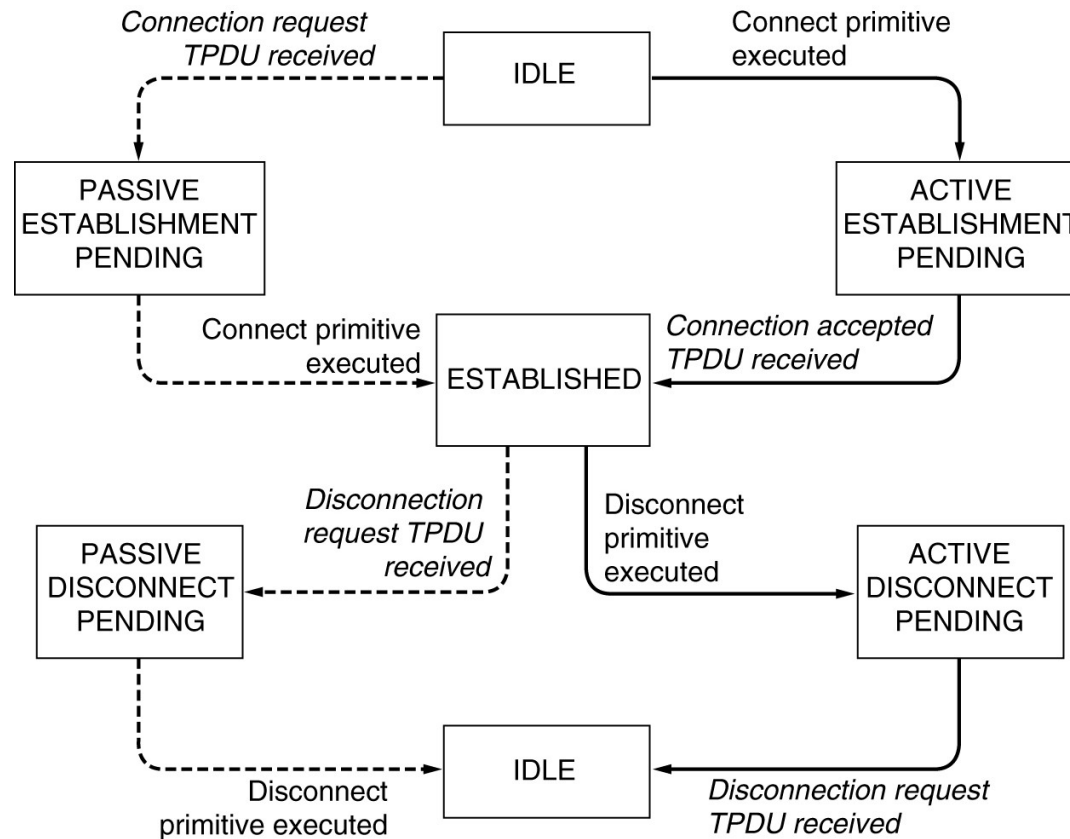
The primitives for a simple transport service.

Transport Service Primitives (2)



The nesting of TPDUs, packets, and frames.

Transport Service Primitives (3)



A state diagram for a simple connection management scheme. Transitions labeled in italics are caused by packet arrivals. The solid lines show the client's state sequence. The dashed lines show the server's state sequence.

Berkeley Sockets

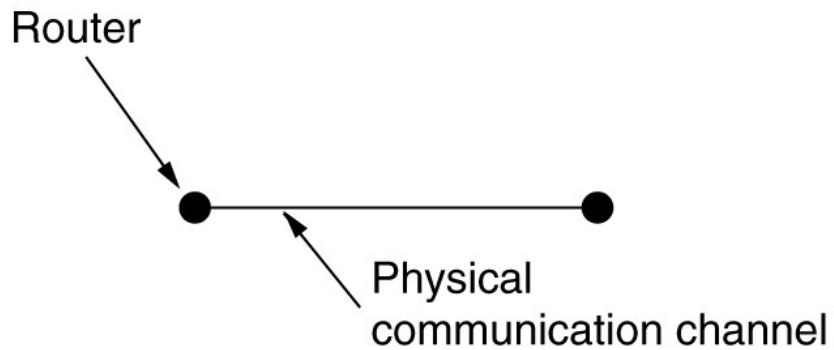
Primitive	Meaning
SOCKET	Create a new communication end point
BIND	Attach a local address to a socket
LISTEN	Announce willingness to accept connections; give queue size
ACCEPT	Block the caller until a connection attempt arrives
CONNECT	Actively attempt to establish a connection
SEND	Send some data over the connection
RECEIVE	Receive some data from the connection
CLOSE	Release the connection

The socket primitives for TCP.

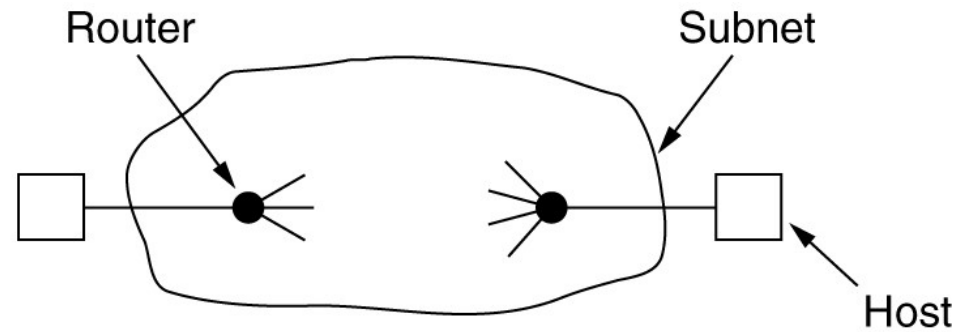
Elements of Transport Protocols

- Addressing
- Connection Establishment
- Connection Release

Transport Protocol



(a)

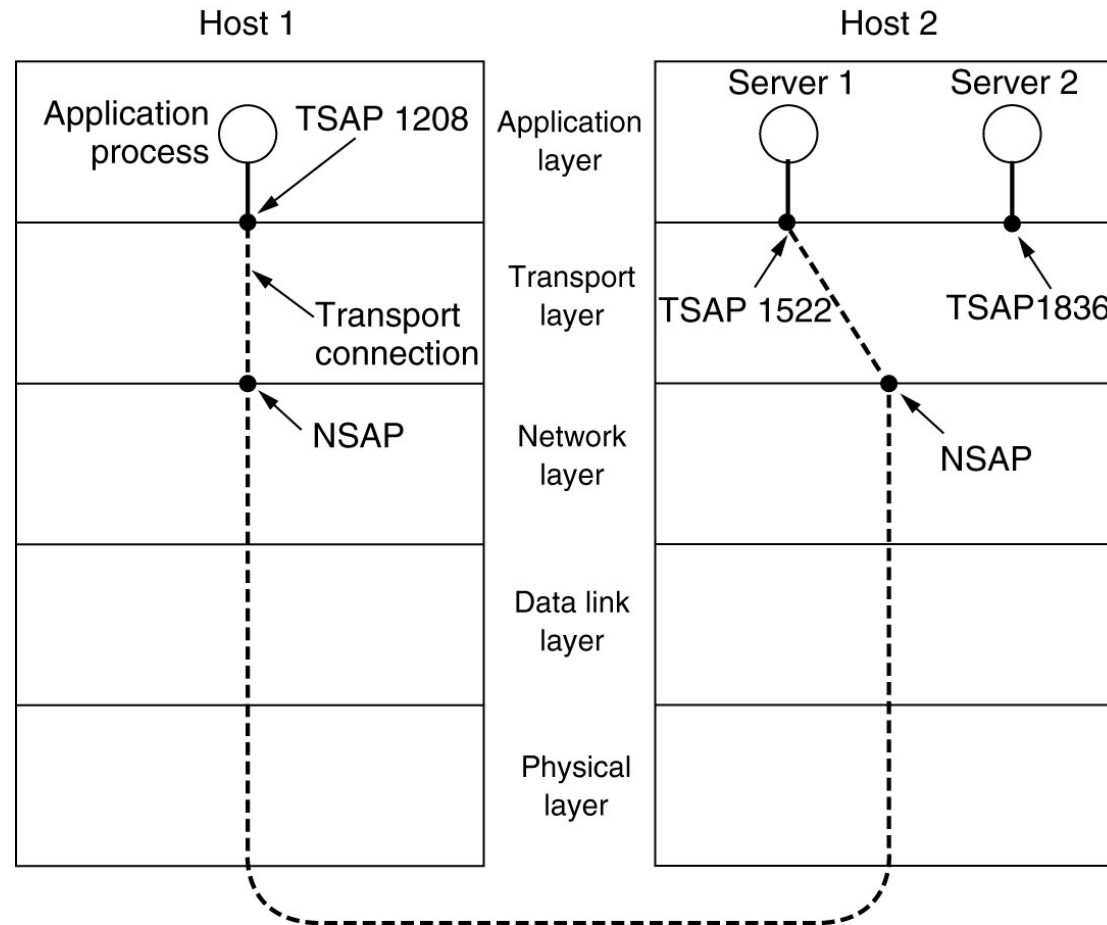


(b)

(a) Environment of the data link layer.

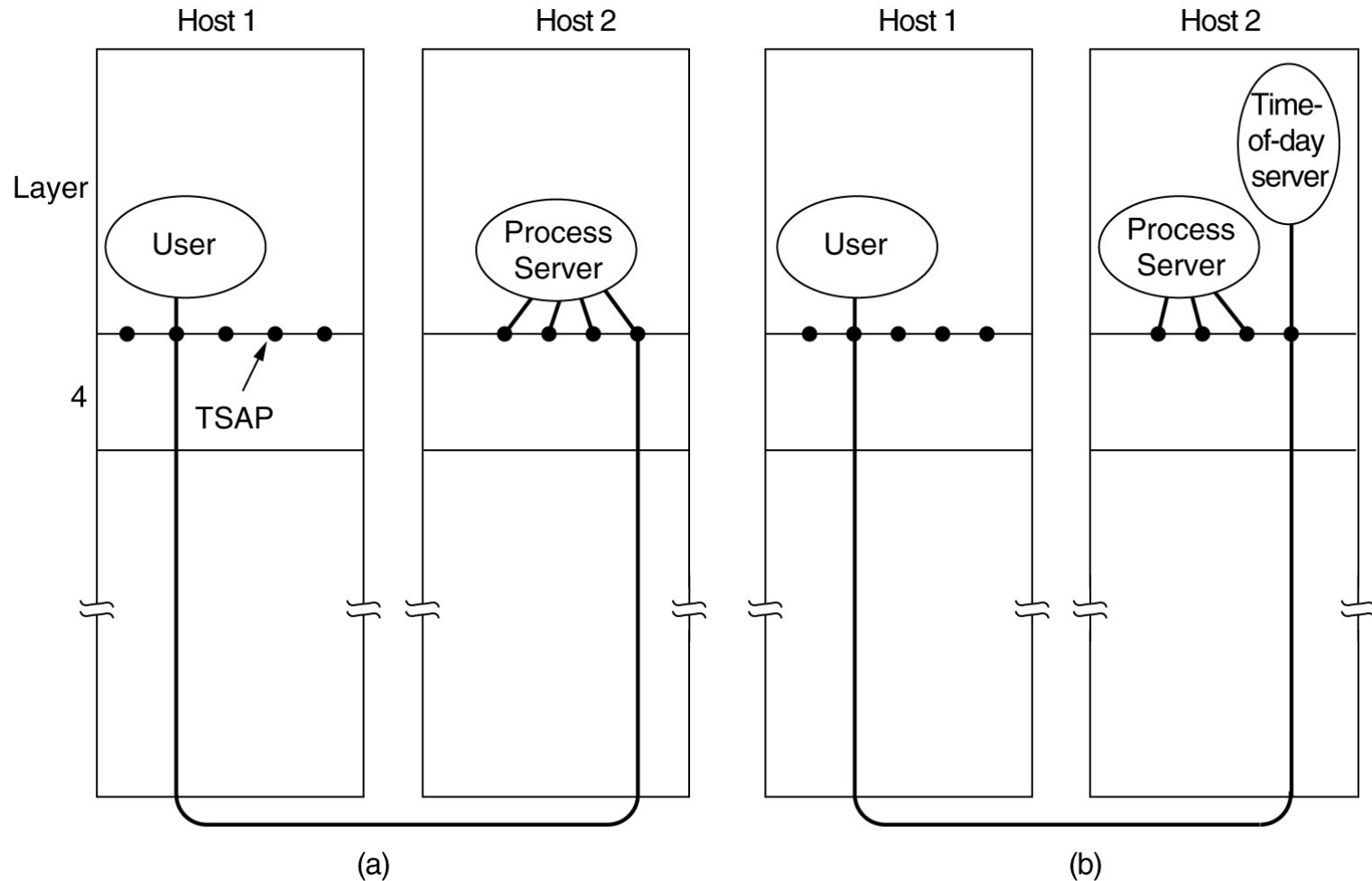
(b) Environment of the transport layer.

Addressing



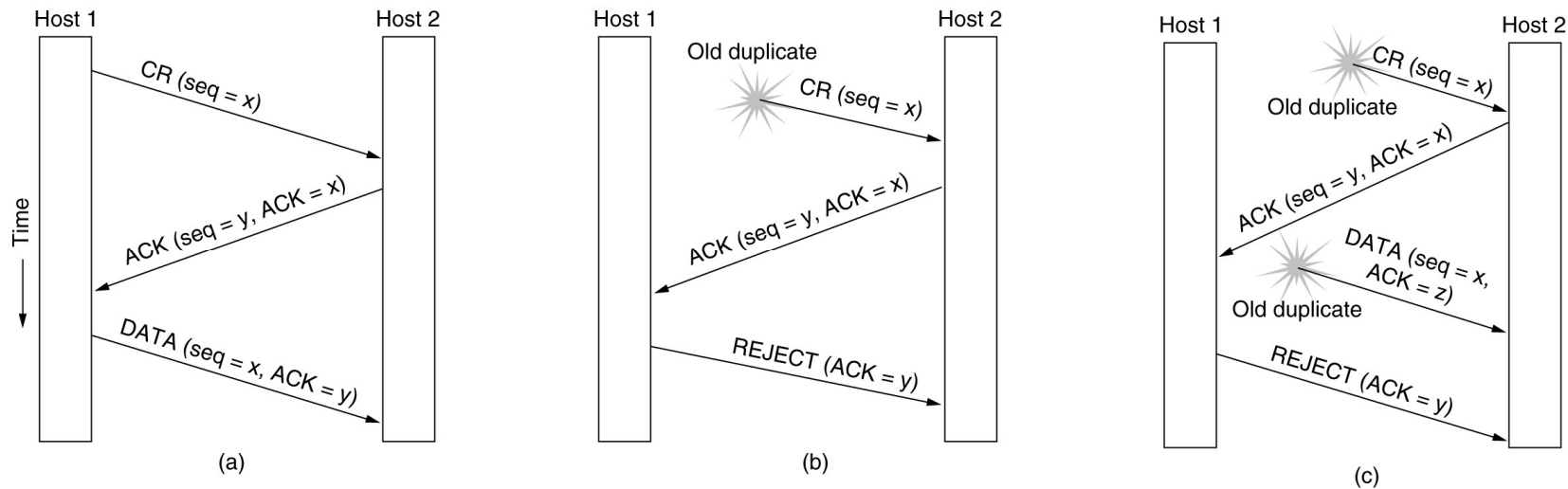
TSAPs, NSAPs and transport connections.

Connection Establishment



How a user process in host 1 establishes a connection with a time-of-day server in host 2.

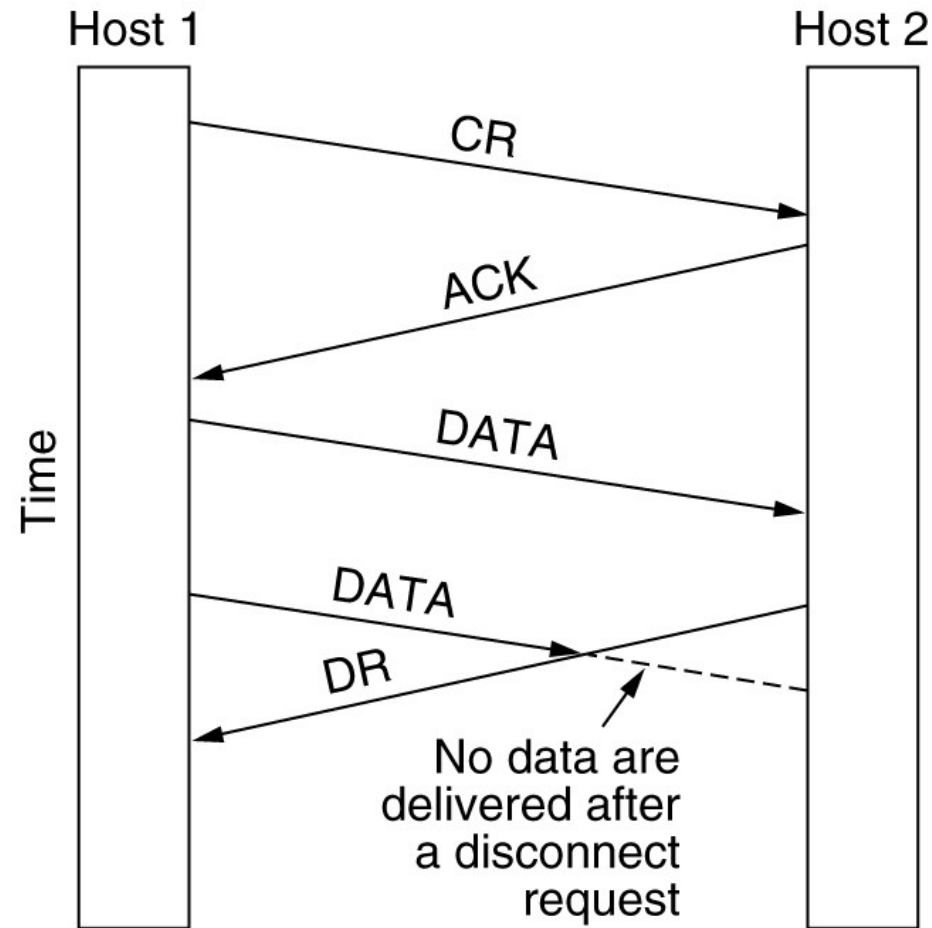
Connection Establishment (3)



Three protocol scenarios for establishing a connection using a three-way handshake. CR denotes CONNECTION REQUEST.

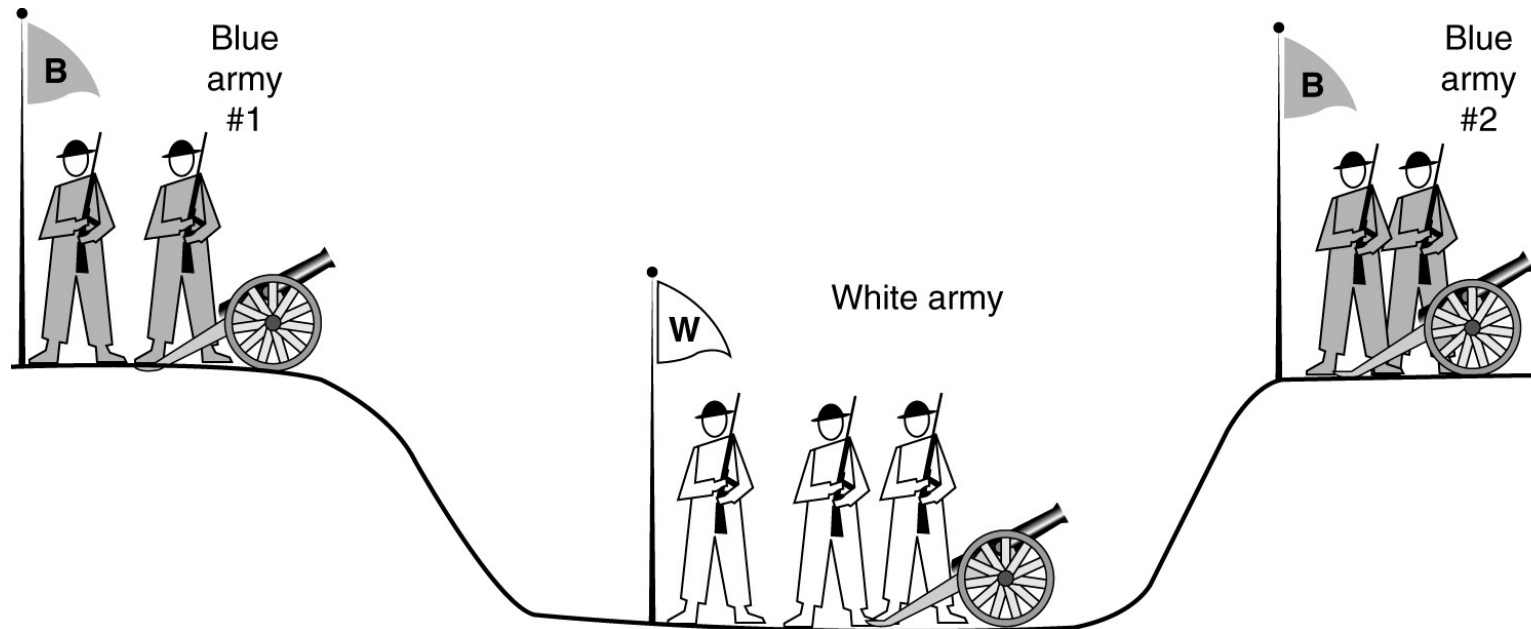
- (a) Normal operation,
- (b) Old CONNECTION REQUEST appearing out of nowhere.
- (c) Duplicate CONNECTION REQUEST and duplicate ACK.

Connection Release



Abrupt disconnection with loss of data.

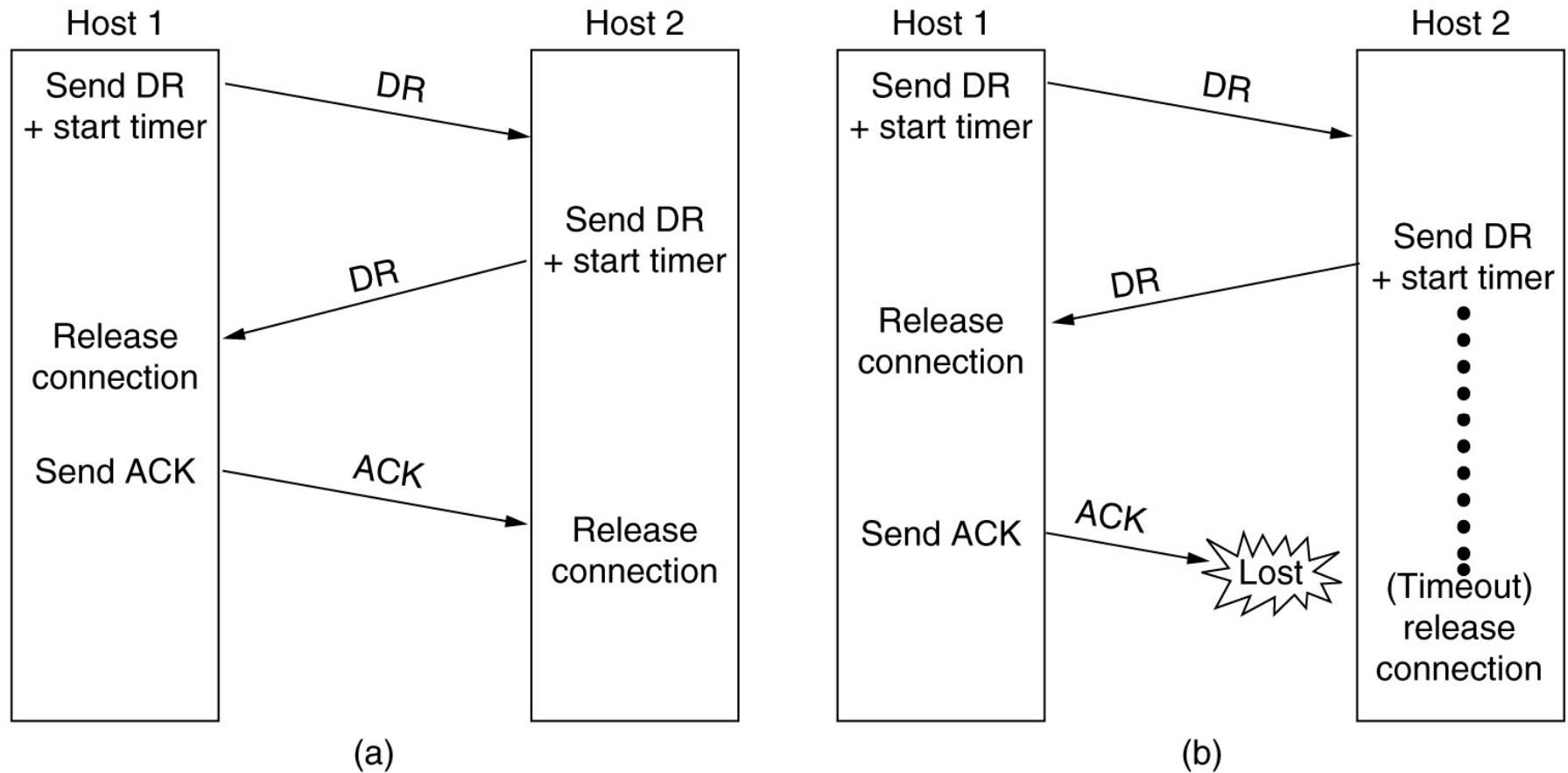
Connection Release (2)



The two-army problem.

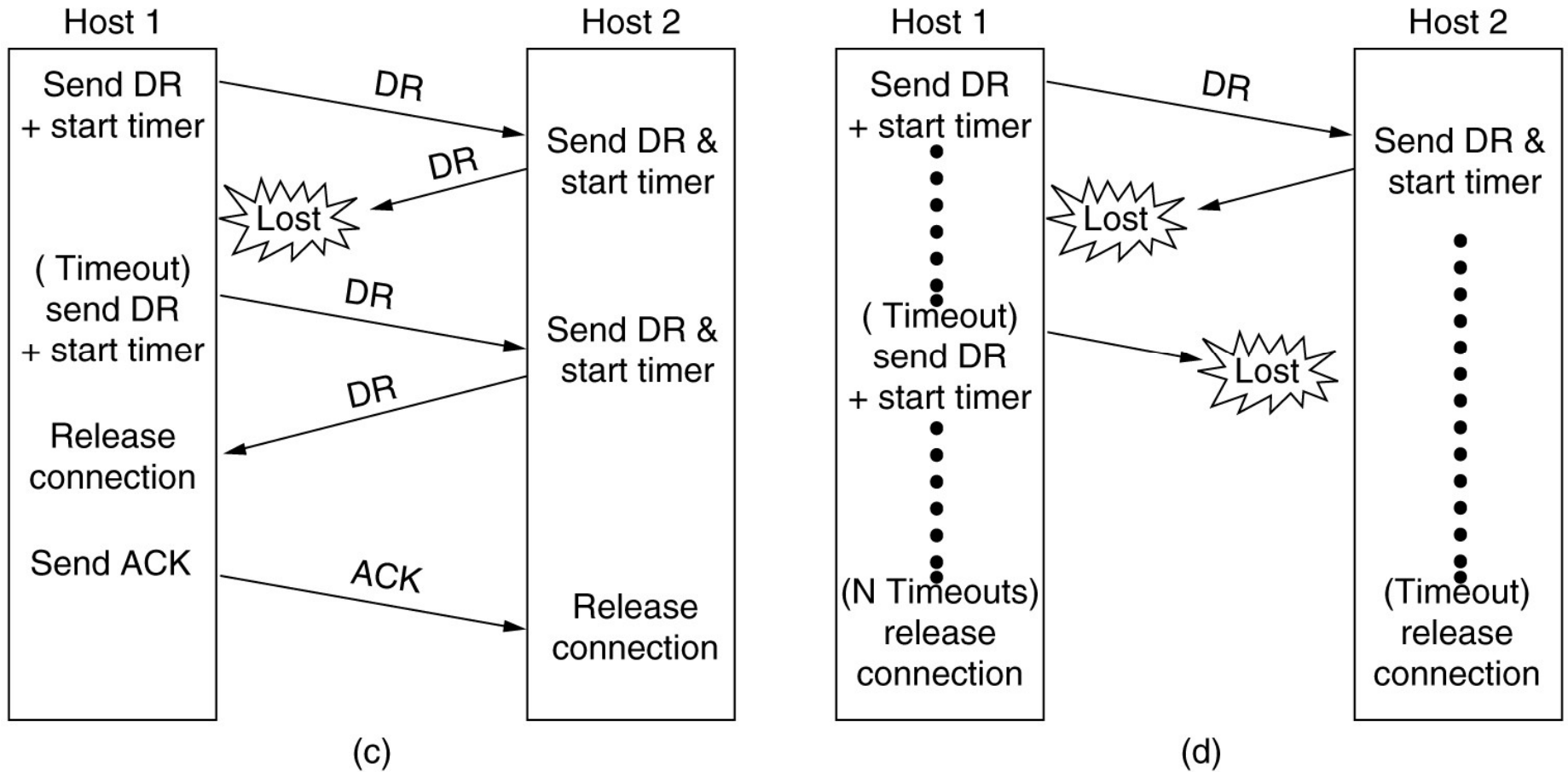
- To avoid two army problem, timers are used. If response to FIN is not forthcoming within two maximum packet lifetimes, the sender of the FIN releases the connection. The other side eventually notice that nobody seems to be listening to it any more and will time out as well.

Connection Release (3)



Four protocol scenarios for releasing a connection. (a) Normal case of a three-way handshake. (b) final ACK lost.

Connection Release (4)



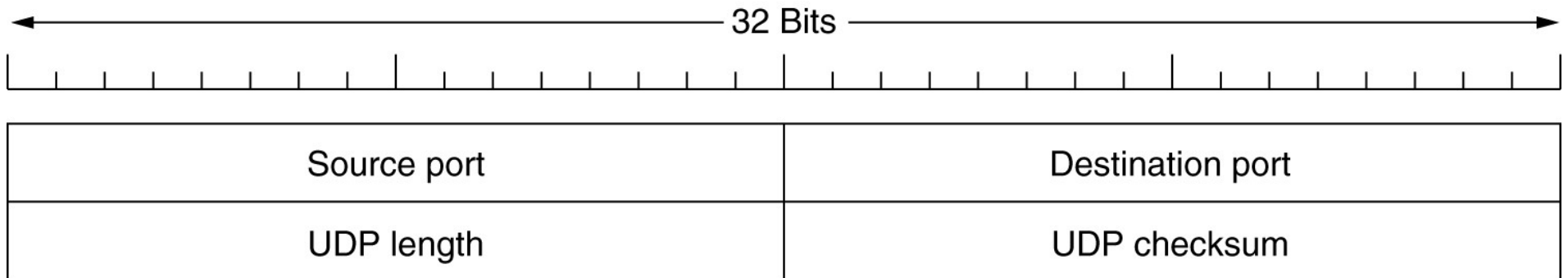
(c) Response lost. (d) Response lost and subsequent DRs lost.

The Internet Transport Protocols: UDP

User Datagram Protocol

- UDP is an **unreliable and connectionless protocol**. So, there is no need to establish a connection prior to data transfer.
- The UDP enables process to process communication.

Introduction to UDP



The UDP header.

Source Port: Source Port is a 2 Byte long field used to identify the port number of the source.

Destination Port: It is a 2 Byte long field, used to identify the port of the destined packet.

Length: Length is the length of UDP including the header and the data. It is a 16-bits field.

Checksum: Checksum is 2 Bytes long field. It is the 16-bit one's complement of the one's complement sum of the UDP header, the pseudo-header of information from the IP header, and the data, padded with zero octets at the end (if necessary) to make a multiple of two octets.

Applications

Used for simple request-response communication when the size of data is less and hence there is lesser concern about flow and error control.

It is a suitable protocol for multicasting as UDP supports packet switching.

UDP is widely used in online gaming

Streaming media applications

VoIP (Voice over Internet Protocol) services, such as Skype and WhatsApp, use UDP for real-time voice communication.

DNS (Domain Name System) also uses UDP for its query/response messages.

The Internet Transport Protocols: TCP

- Introduction to TCP
- The TCP Service Model
- The TCP Protocol
- The TCP Segment Header
- TCP Connection Establishment
- TCP Connection Release

TCP (Transmission Control Protocol)

Key Features:

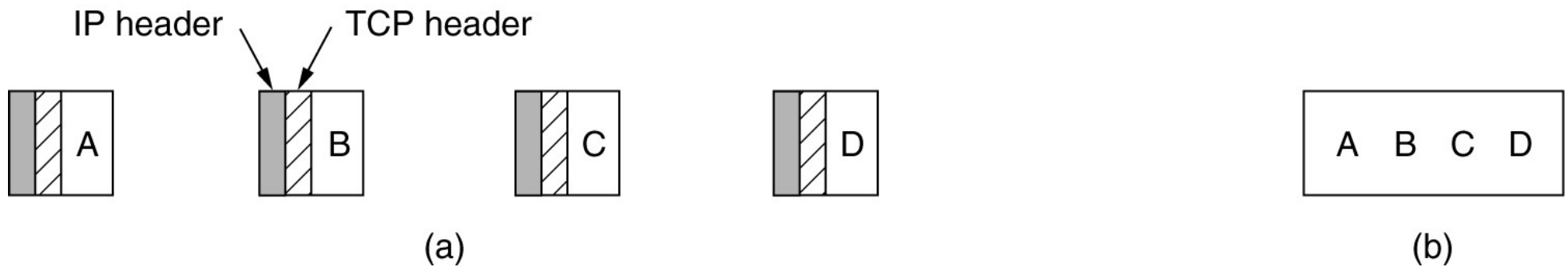
- **Connection-oriented** – a connection is established before data transfer.
- **Reliable** – ensures data is delivered, in order, without loss or duplication.
- **Slower** – because of acknowledgments, retransmissions, and congestion control.
- **Heavyweight** – more overhead.
- **Use Cases:**
 - Web browsing (HTTP/HTTPS)
 - Email (SMTP)
 - File transfer (FTP)
 - Secure shell (SSH)

The TCP Service Model

Port	Protocol	Use
21	FTP	File transfer
23	Telnet	Remote login
25	SMTP	E-mail
69	TFTP	Trivial File Transfer Protocol
79	Finger	Lookup info about a user
80	HTTP	World Wide Web
110	POP-3	Remote e-mail access
119	NNTP	USENET news

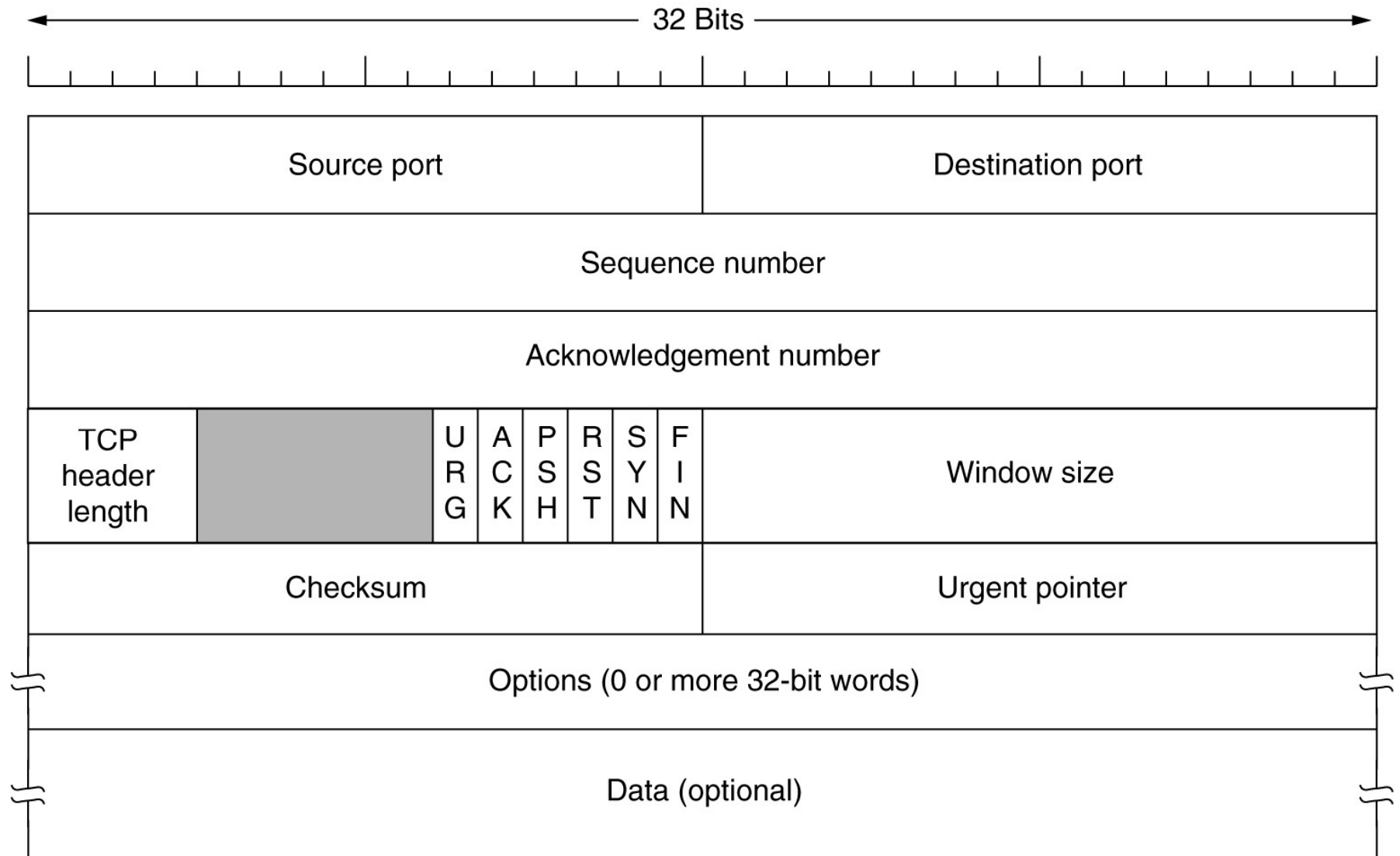
Some assigned ports.

The TCP Service Model (2)



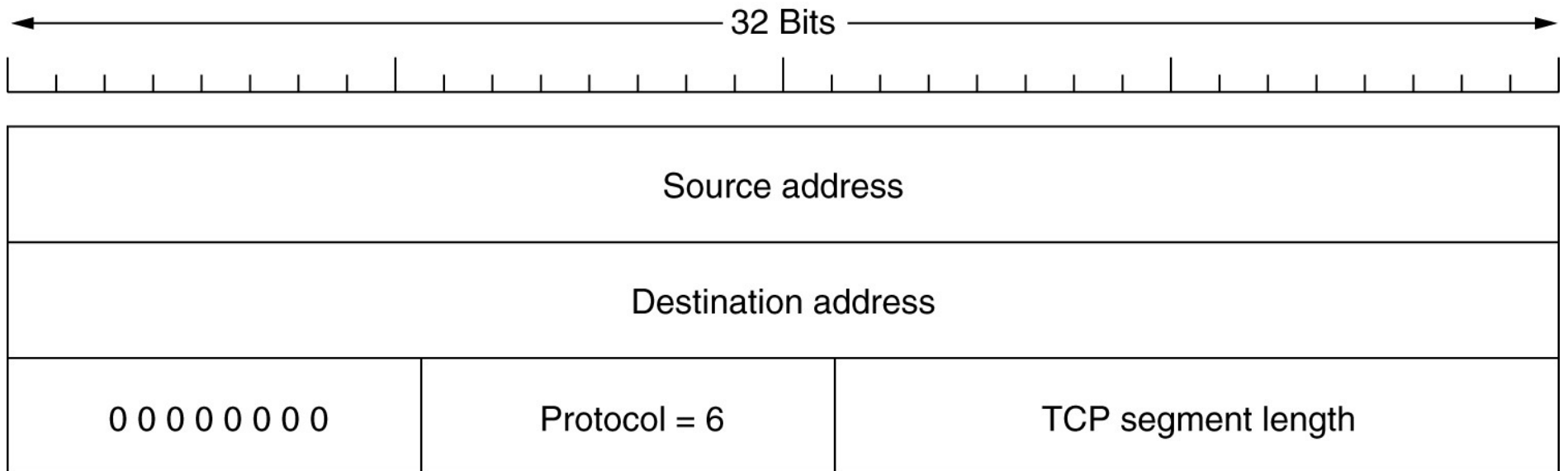
- (a) Four 512-byte segments sent as separate IP datagrams.
- (b) The 2048 bytes of data delivered to the application in a single READ CALL.

The TCP Segment Header



TCP Header.

The TCP Segment Header (2)



The pseudoheader included in the TCP checksum.

The TCP Segment Header

Every TCP segment consists of a 20 byte fixed format header. Header options may follow the fixed header. With a header so that it can tag up to 65535 data bytes.

Source Port

It is a 16-bit source port number used by the receiver to reply.

Destination Port

It is a 16-bit destination port number.

Sequence Number

The sequence number of the first data byte in this segment. During the SYN Control bit is set, and the sequence number is n , and the first data byte is $n + 1$.

Acknowledgement Number

If the ACK control bit is set, this field contains the next number that the receiver expects to receive.

Data Offset

The several 32-bit words in the TCP header shows from where the user data begins.

Reserved (6 bit)

It is reserved for future use.

URG

It indicates an urgent pointer field that data type is urgent or not.

ACK

It indicates that the acknowledgement field in a segment is significant, as discussed early.

PUSH

The PUSH flag is set or reset according to a data type that is sent immediately or not.

RST

It Resets the connection.

SYN

It synchronizes the sequence number.

FIN

This indicates no more data from the sender.

Window

It is used in Acknowledgement segment. It specifies the number of data bytes, beginning with the one indicated in the acknowledgement number field that the receiver is ready to accept.

Checksum

It is used for error detection.

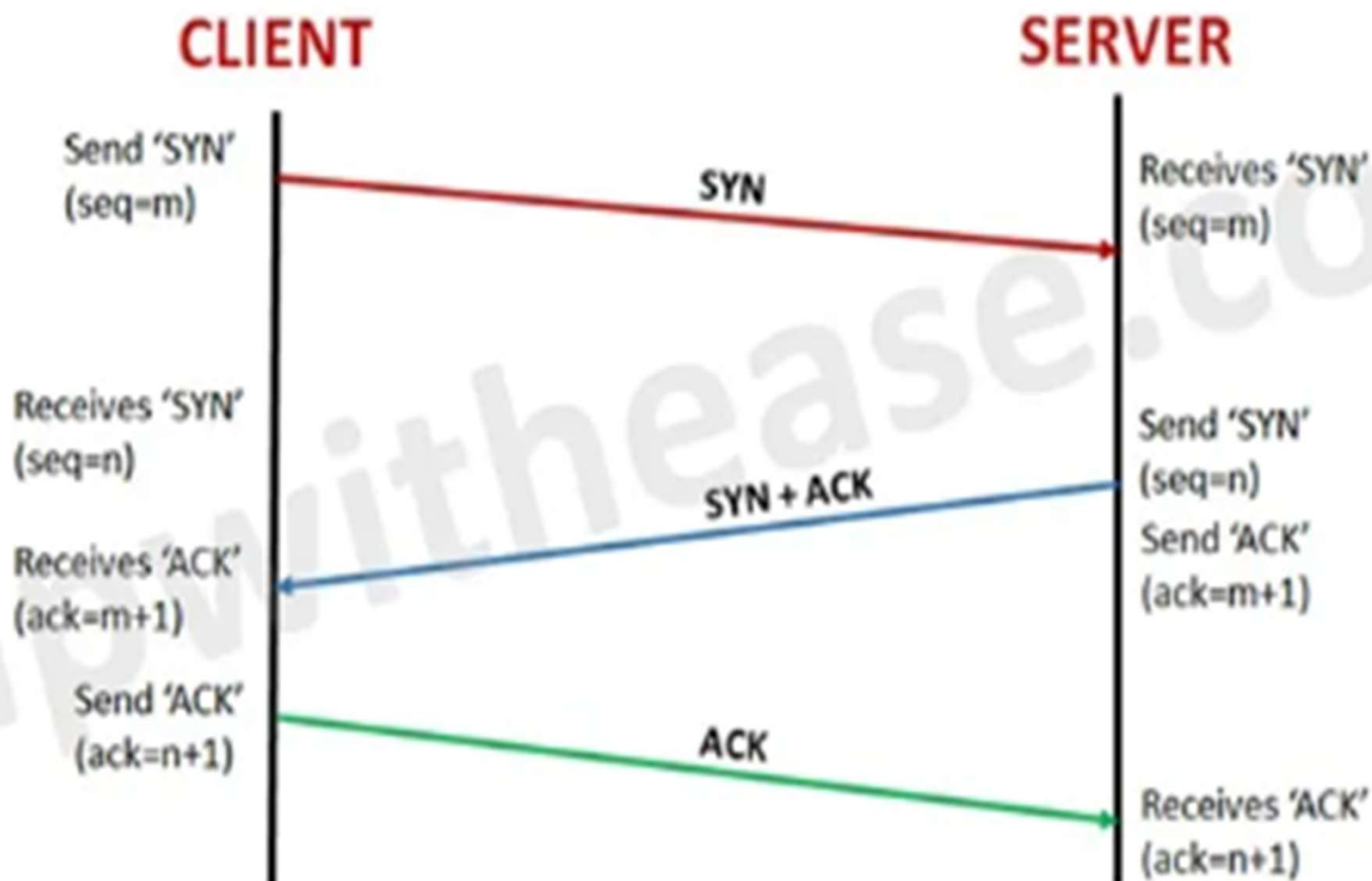
Options

The IP datagram options provide additional punctuality. It can use several optional parameters between a TCP sender and receiver. It depends on the options used. The length of the field may vary in size, but it can't be larger than 40 bytes due to the header field's size, which is 4 bit.

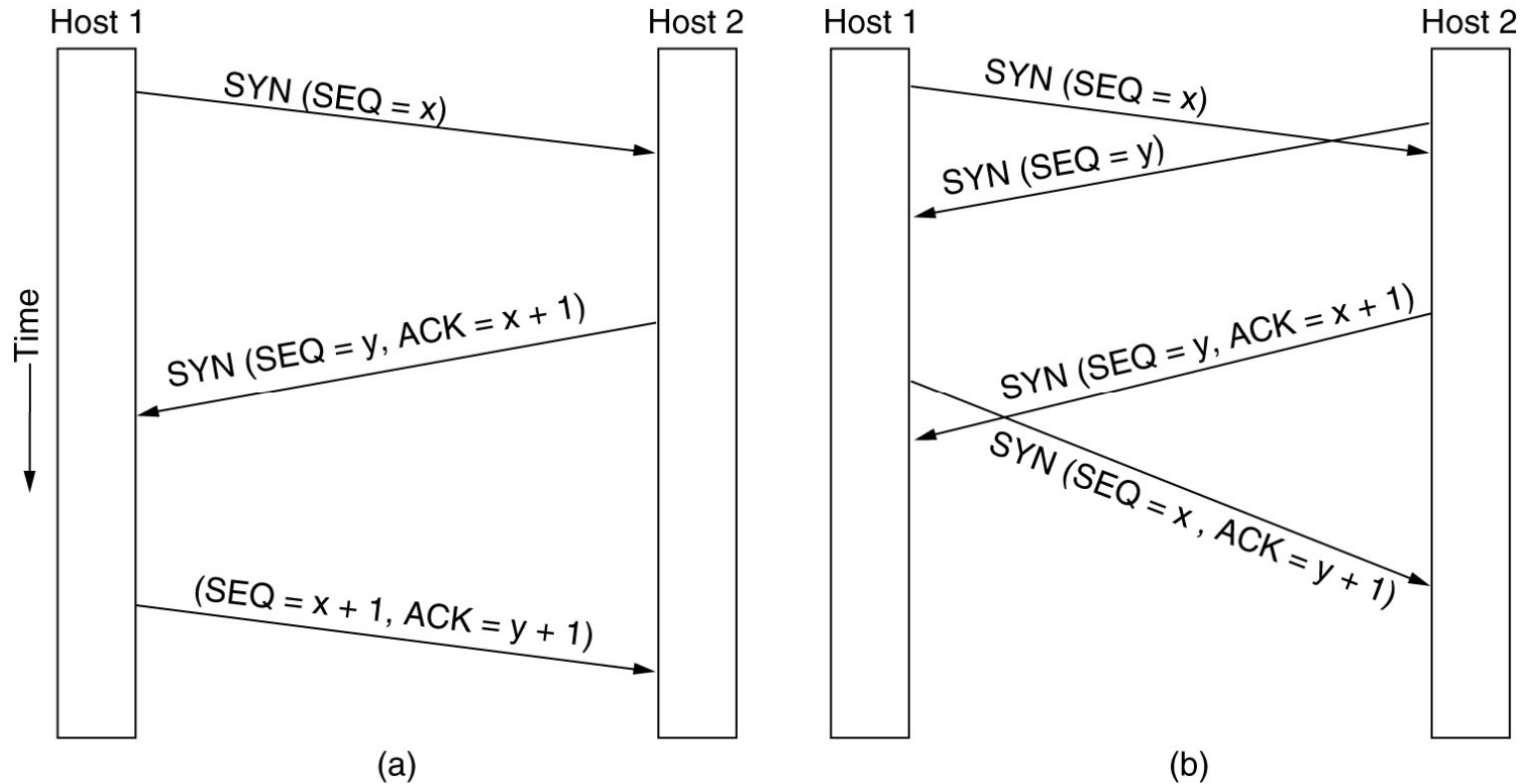
The most typical option is the maximum segment size MASS option. A TCP receiver communicates to the TCP sender the total length of the segment it can accept with this option.

TCP 3-Way handshake Process

(Connection Establishment)



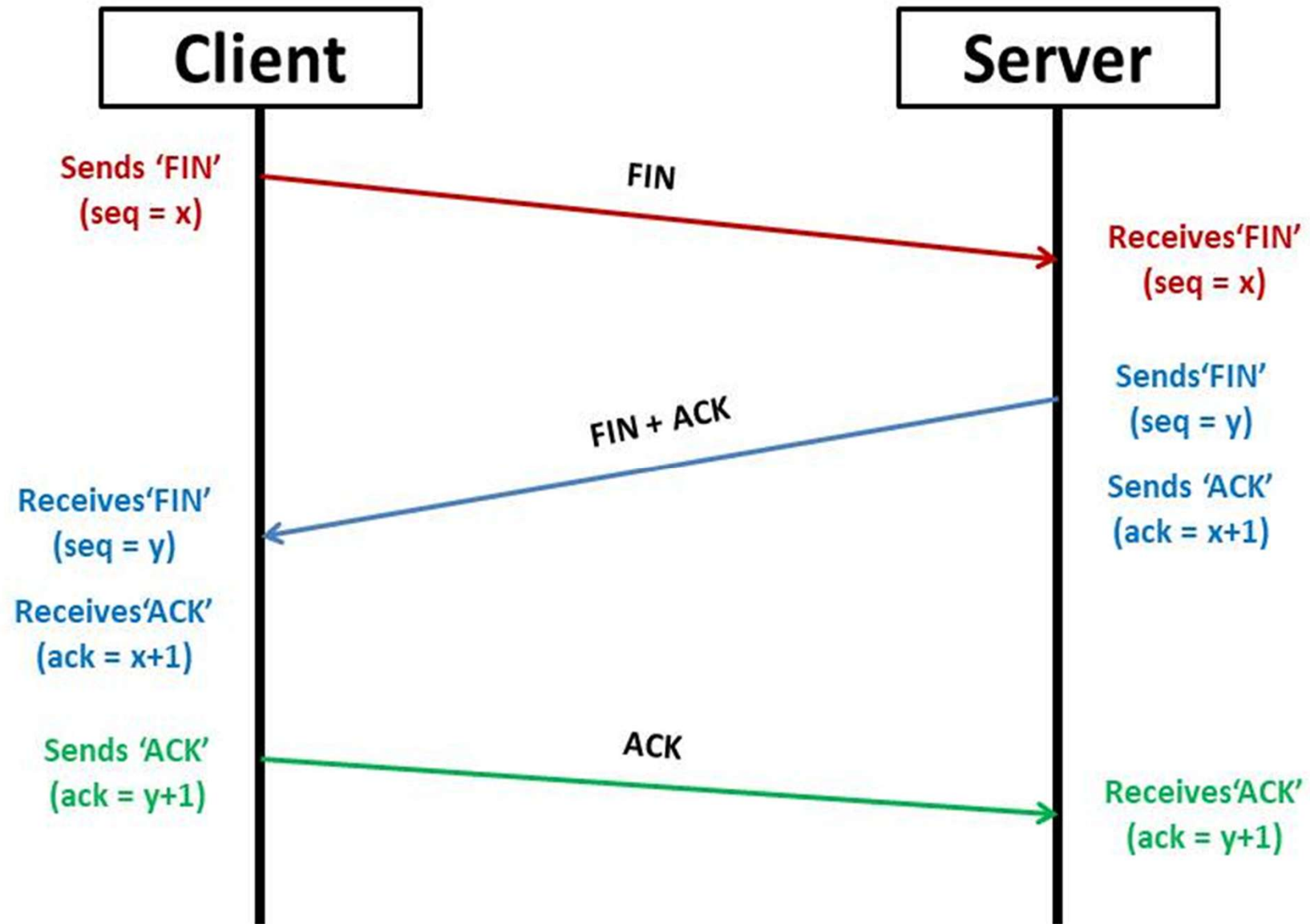
TCP Connection Establishment



- (a) TCP connection establishment in the normal case.
- (b) Call collision.

TCP Connection Release

- Although TCP connections are full duplex, to understand connection release it is best to think them as a pair of simplex connections.
- To release a connection, either party can send a TCP segment with the FIN bit set, which means that it has no more data to transmit.
- When FIN bit is acknowledged, that direction is shut down for new data. Data may continue to flow indefinitely in the other direction, however.
- When both directions have been shut down, the connection is released.
- Normally four TCP segments are needed, but it is possible for first ACK and second FIN to be contained in same segment, reducing the total count to three.



3-Way Handshaking(for terminating connection)

The Application Layer

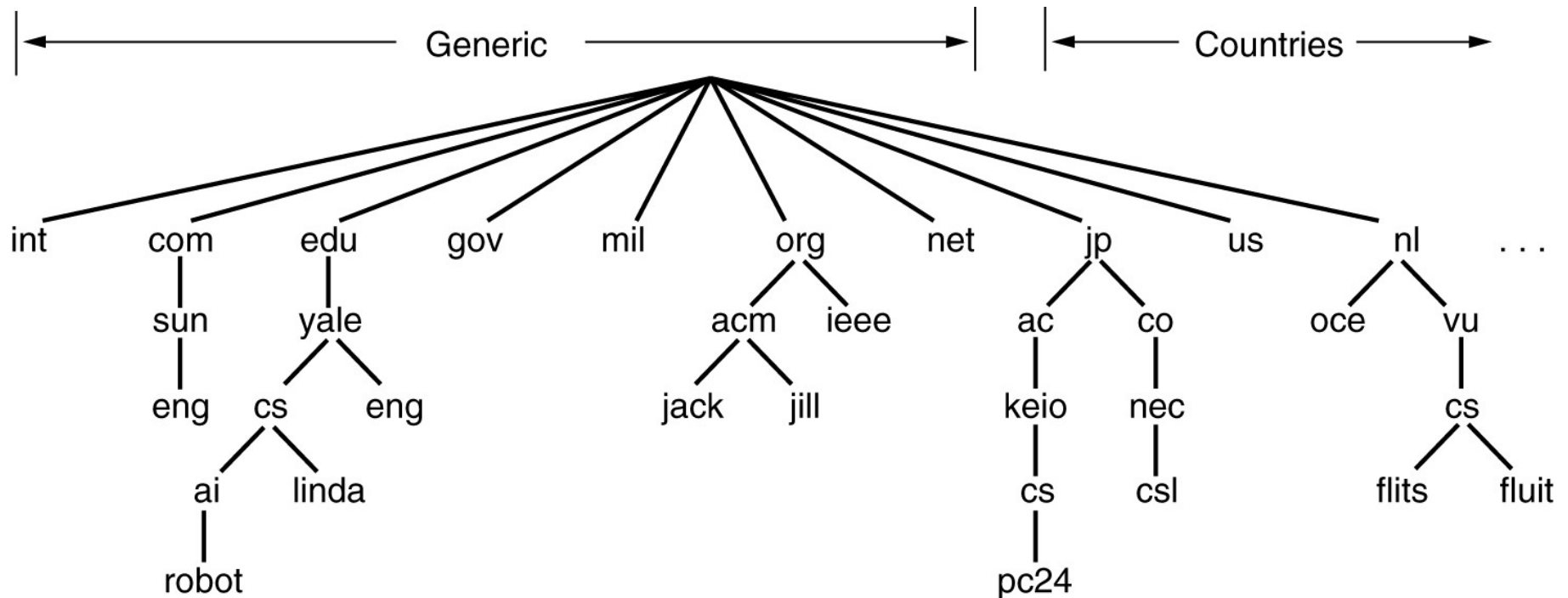
DNS – The Domain Name System

- The DNS Name Space
- Resource Records
- Name Servers

DNS

- DNS is the invention of a hierarchical, domain-based naming scheme and a distributed database system for implementing this naming scheme.
- DNS is used to map a name onto an IP address, an application program calls a library procedure called the Resolver, passing it the name as a parameter.
- The Resolver sends a UDP packet to a local DNS server, which then looks up the name and returns IP address, the program can then establish a TCP connection with the destination or send it UDP packets.

The DNS Name Space



A portion of the Internet domain name space.

DNS Name Space

- Internet is divided into 200 top-level domains, where each domain covers many hosts.
- Each domain is partitioned into subdomains, and these are further partitioned, and so on.
- Top-level domains : Generic
Countries.
- Generic Domains were com(commercial), edu(educational institutions), gov, int(international organizations), mil, net(network providers), and org(non profit organization).
- The Country domains include one entry for every country.
- Each domain is named by the path upward from it the (unnamed) root.

DNS(contd..)

- Domain names can be either absolute or relative.
- An absolute domain name always ends with a period (e.g.,*eng.sun.com*)
- Relative does not.
- Domain names are case insensitive.
- Domains can be inserted into the tree in two different ways.
- Each domain controls how it allocates the domains under it.
- To create a new domain, permission is required of the domain in which it will be included. So that name conflicts are avoided.
- Once domain has been created & registered, it can create subdomains, without getting permission from anybody higher up the tree.

Resource Records

- Every Domain, whether it is a single host or a top-level domain, can have a set of resource records associated with it.
- For single host, the most common resource record is just its IP address, but many other kinds of resource records also exist.
- When a resolver gives a domain name to DNS, it gets back resource records associated with that name. Thus, the primary function of DNS is to map domain names onto resource records.
- A resource record is a five-tuple. The format is as follows:

Domain_name	Time_to_live	Class	Type	Value
-------------	--------------	-------	------	-------
- **Domain_name** :tells the domain to which this record applies
- **Time_to_live** field gives an indication of how stable the record is.
- **Class**: For Internet information, it is always IN
 - For non-Internet information, other codes can be used.

Resource Records

Type: This field tells what kind of record this is. The important types are listed below

Type	Meaning	Value
SOA	Start of Authority	Parameters for this zone
A	IP address of a host	32-Bit integer
MX	Mail exchange	Priority, domain willing to accept e-mail
NS	Name Server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
HINFO	Host description	CPU and OS in ASCII
TXT	Text	Uninterpreted ASCII text

The principal DNS resource records types.

Value: This can be a number, a domain name, or an ASCII string.

Resource Records (2)

```
; Authoritative data for cs.vu.nl
cs.vu.nl.      86400  IN  SOA      star boss (952771,7200,7200,2419200,86400)
cs.vu.nl.      86400  IN  TXT      "Divisie Wiskunde en Informatica."
cs.vu.nl.      86400  IN  TXT      "Vrije Universiteit Amsterdam."
cs.vu.nl.      86400  IN  MX       1 zephyr.cs.vu.nl.
cs.vu.nl.      86400  IN  MX       2 top.cs.vu.nl.

flits.cs.vu.nl. 86400  IN  HINFO    Sun Unix
flits.cs.vu.nl. 86400  IN  A        130.37.16.112
flits.cs.vu.nl. 86400  IN  A        192.31.231.165
flits.cs.vu.nl. 86400  IN  MX       1 flits.cs.vu.nl.
flits.cs.vu.nl. 86400  IN  MX       2 zephyr.cs.vu.nl.
flits.cs.vu.nl. 86400  IN  MX       3 top.cs.vu.nl.
www.cs.vu.nl.   86400  IN  CNAME    star.cs.vu.nl
ftp.cs.vu.nl.   86400  IN  CNAME    zephyr.cs.vu.nl

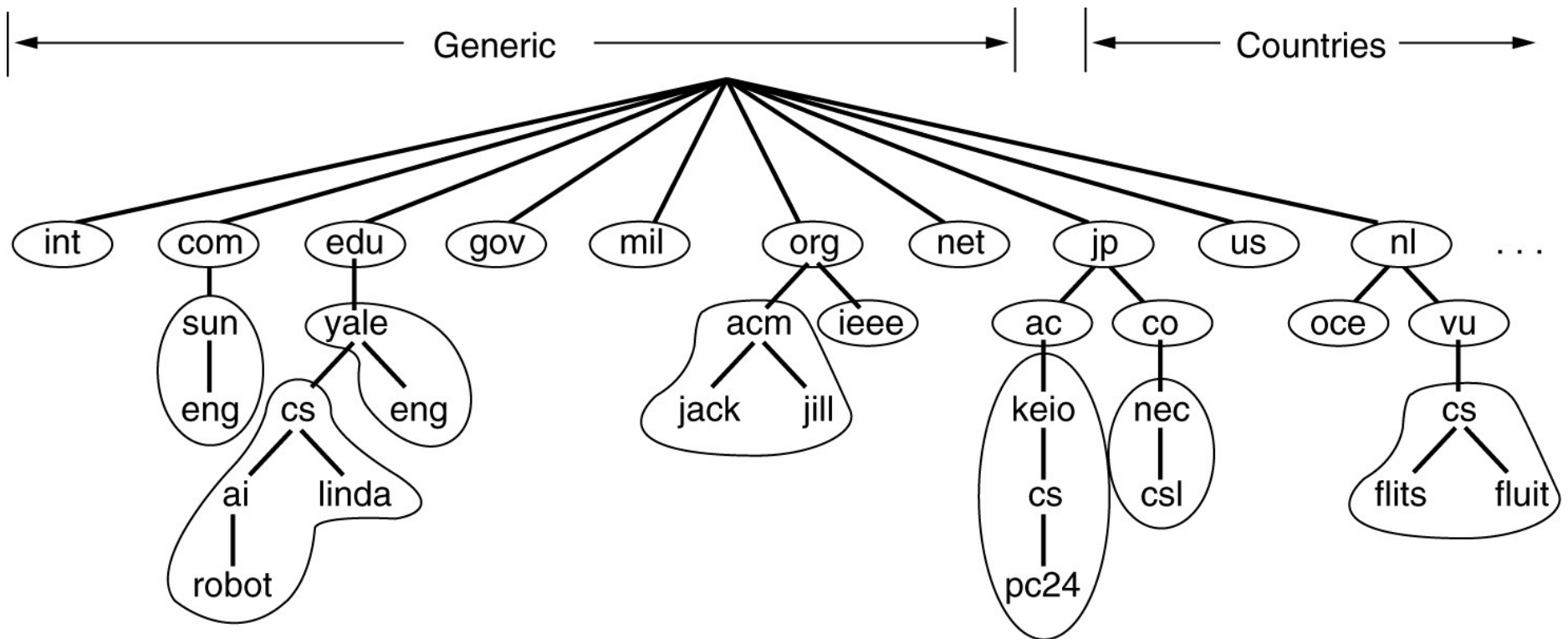
rowboat                IN  A        130.37.56.201
                        IN  MX       1 rowboat
                        IN  MX       2 zephyr
                        IN  HINFO    Sun Unix

little-sister          IN  A        130.37.62.23
                        IN  HINFO    Mac MacOS

laserjet                IN  A        192.31.231.216
                        IN  HINFO    "HP Laserjet IIISi" Proprietary
```

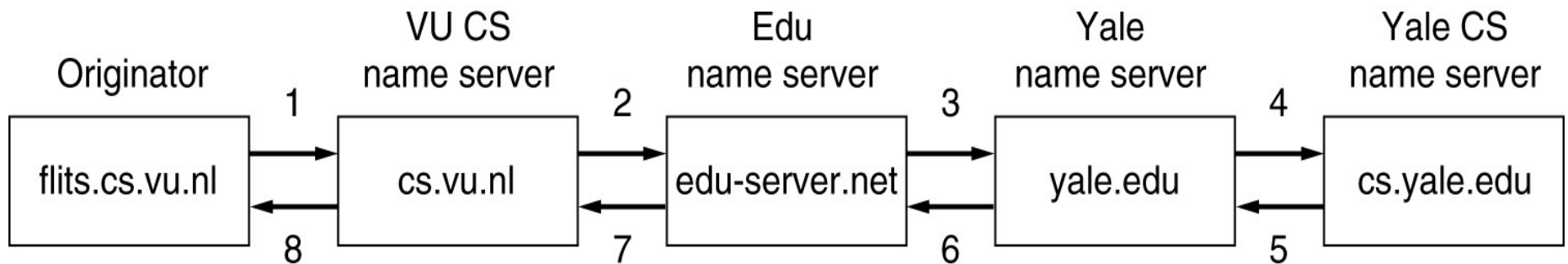
A portion of a possible DNS database for *cs.vu.nl*.

Name Servers



Part of the DNS name space showing the division into zones.

Name Servers (2)



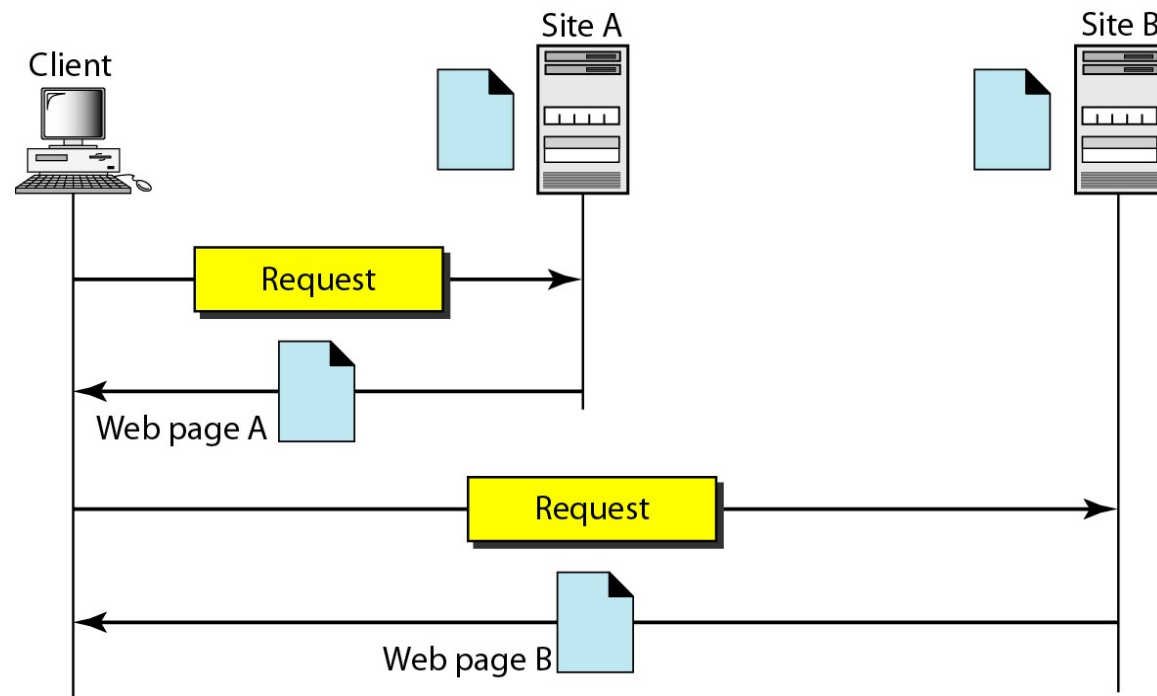
How a resolver looks up a remote name in eight steps.

The World Wide Web

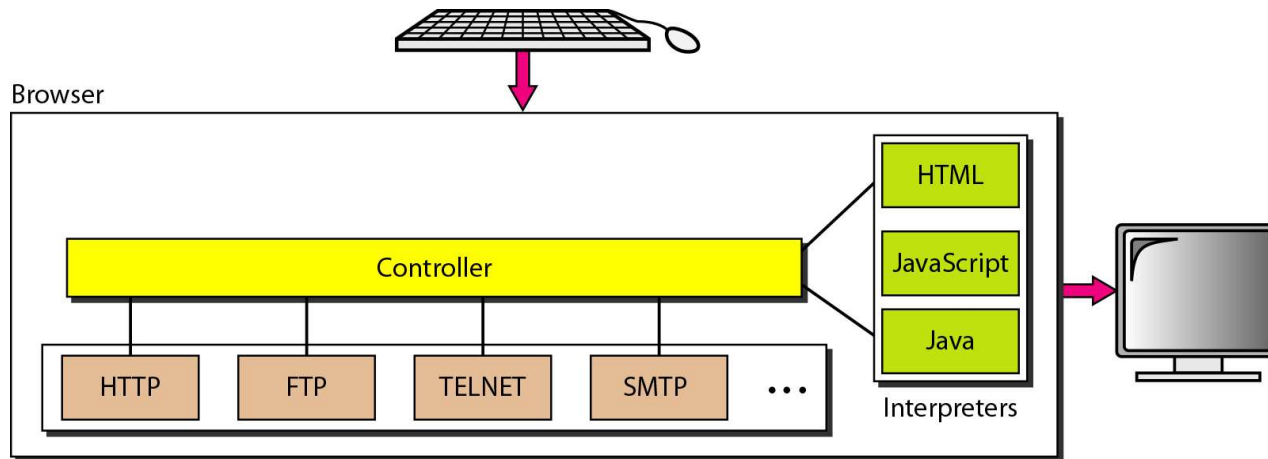
- Architectural Overview
- Static Web Documents
- Dynamic Web Documents
- HTTP – The HyperText Transfer Protocol
- Performance Enhancements
- The Wireless Web

Architecture of WWW

The WWW today is a distributed client/server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called sites.



Browser and URL



Uniform Resource Locator



•

URLs – Uniform Resource Locaters

Name	Used for	Example
http	Hypertext (HTML)	http://www.cs.vu.nl/~ast/
ftp	FTP	ftp://ftp.cs.vu.nl/pub/minix/README
file	Local file	file:///usr/suzanne/prog.c
news	Newsgroup	news:comp.os.minix
news	News article	news:AA0134223112@cs.utah.edu
gopher	Gopher	gopher://gopher.tc.umn.edu/11/Libraries
mailto	Sending e-mail	mailto:JohnUser@acm.org
telnet	Remote login	telnet://www.w3.org:80

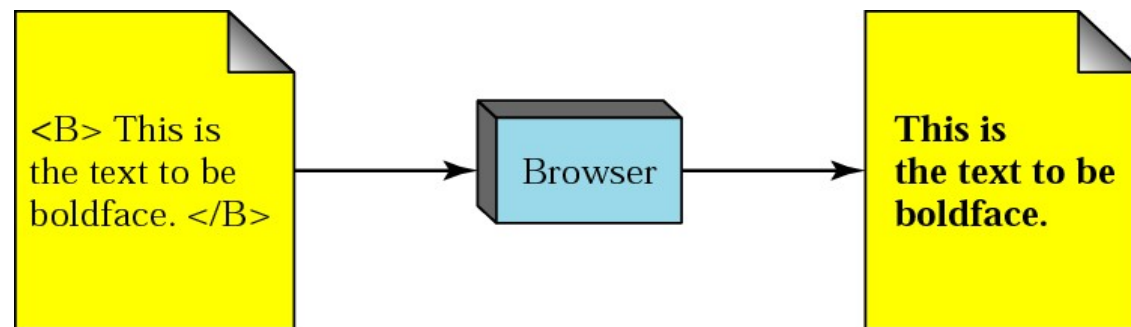
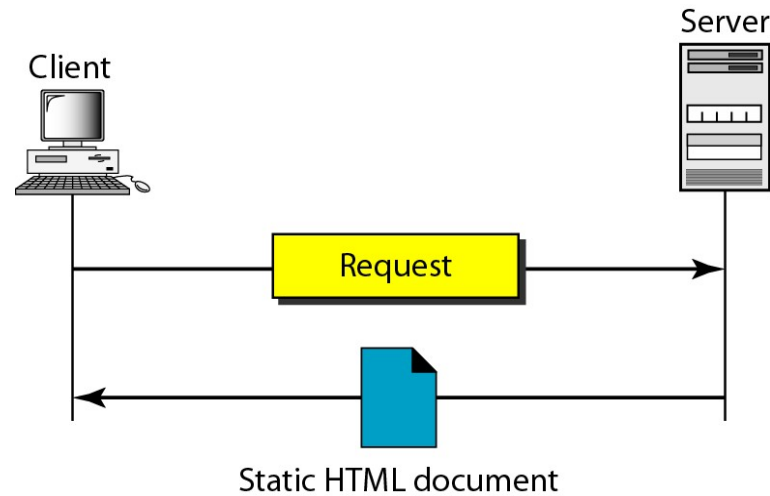
Some common URLs.

Statelessness and Cookies

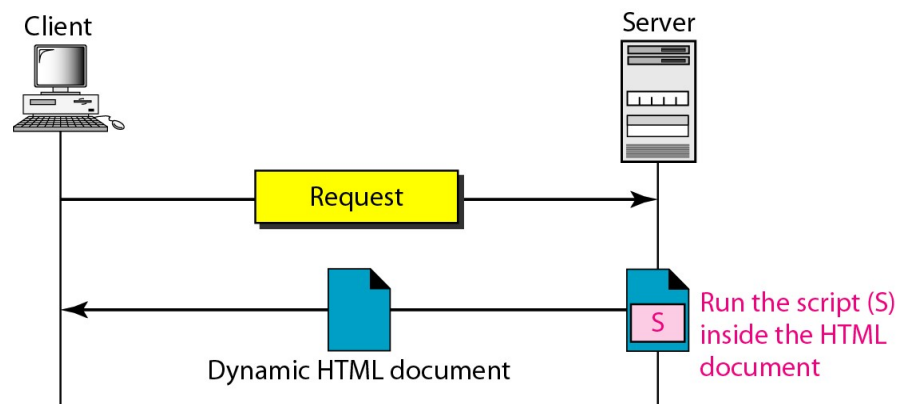
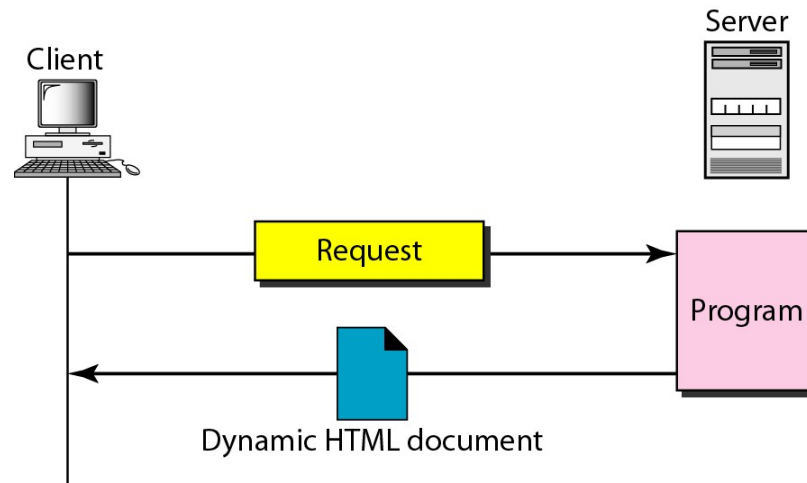
Domain	Path	Content	Expires	Secure
toms-casino.com	/	CustomerID=497793521	15-10-02 17:00	Yes
joes-store.com	/	Cart=1-00501;1-07031;2-13721	11-10-02 14:22	No
aportal.com	/	Prefs=Stk:SUNW+ORCL;Spt:Jets	31-12-10 23:59	No
sneaky.com	/	UserID=3627239101	31-12-12 23:59	No

Some examples of cookies.

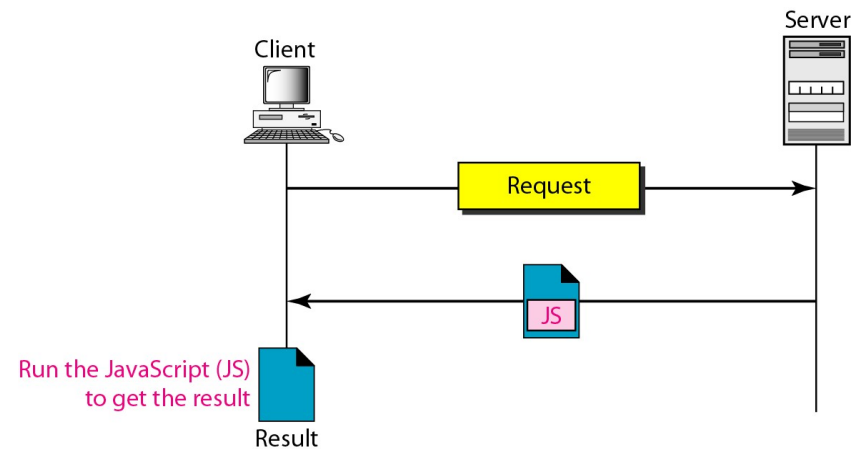
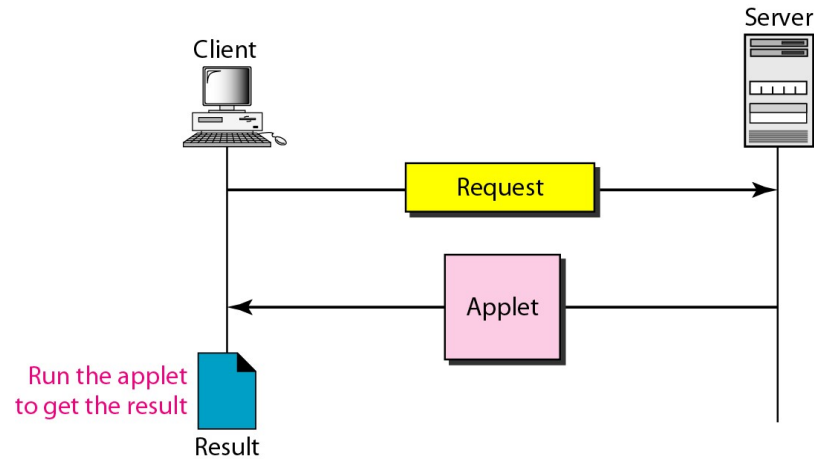
Static Documents: HTML



Dynamic Documents: CGI and Scripting



Active Documents: Java Applets and JavaScript



Dynamic Web Documents (2)

```
<html>
<body>

<h2> This is what I know about you </h2>
<?php echo $HTTP_USER_AGENT ?>

</body>
</html>
```

A sample HTML page with embedded PHP.

Dynamic Web Documents (3)

```
<html>
<body>
<form action="action.php" method="post">
<p> Please enter your name: <input type="text" name="name"> </p>
<p> Please enter your age: <input type="text" name="age"> </p>
<input type="submit">
</form>
</body>
</html>
```

(a)

```
<html>
<body>
<h1> Reply: </h1>
Hello <?php echo $name; ?>.
Prediction: next year you will be <?php echo $age + 1; ?>
</body>
</html>
```

(b)

```
<html>
<body>
<h1> Reply: </h1>
Hello Barbara.
Prediction: next year you will be 25
</body>
</html>
```

(c)

(a) A Web page containing a form. (b) A PHP script for handling the output of the form. (c) Output from the PHP script when the inputs are "Barbara" and 24 respectively.

Client-Side Dynamic Web Page Generation

```
<head>
<script language="javascript" type="text/javascript">
function response(test form) {
    var person = test form.name.value;
    var years = eval(test form.age.value) + 1;
    document.open();
    document.writeln("<html> <body>");
    document.writeln("Hello " + person + ".<br>");
    document.writeln("Prediction: next year you will be " + years + ".");
    document.writeln("</body> </html>");
    document.close();
}
</script>
</head>
```

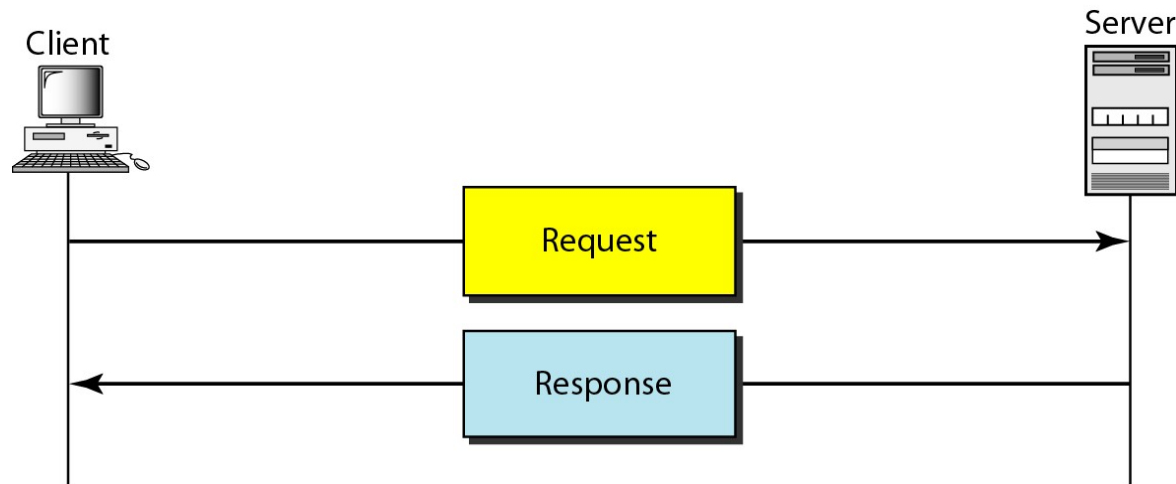
Use of JavaScript
for processing a
form.

```
<body>
<form>
Please enter your name: <input type="text" name="name">
<p>
Please enter your age: <input type="text" name="age">
<p>
<input type="button" value="submit" onclick="response(this.form)">
</form>
</body>
</html>
```

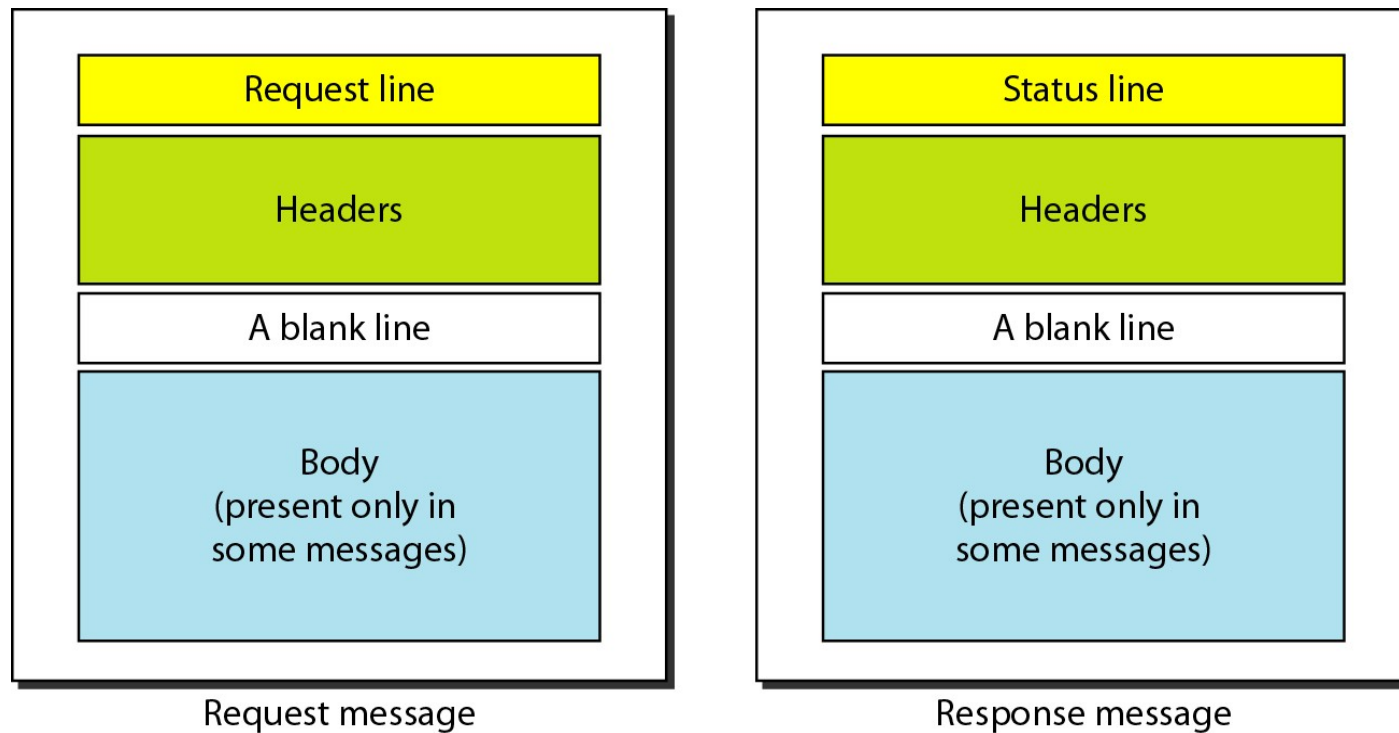

Hypertext Transfer Protocol (HTTP)

HTTP is used mainly to access data on the WWW

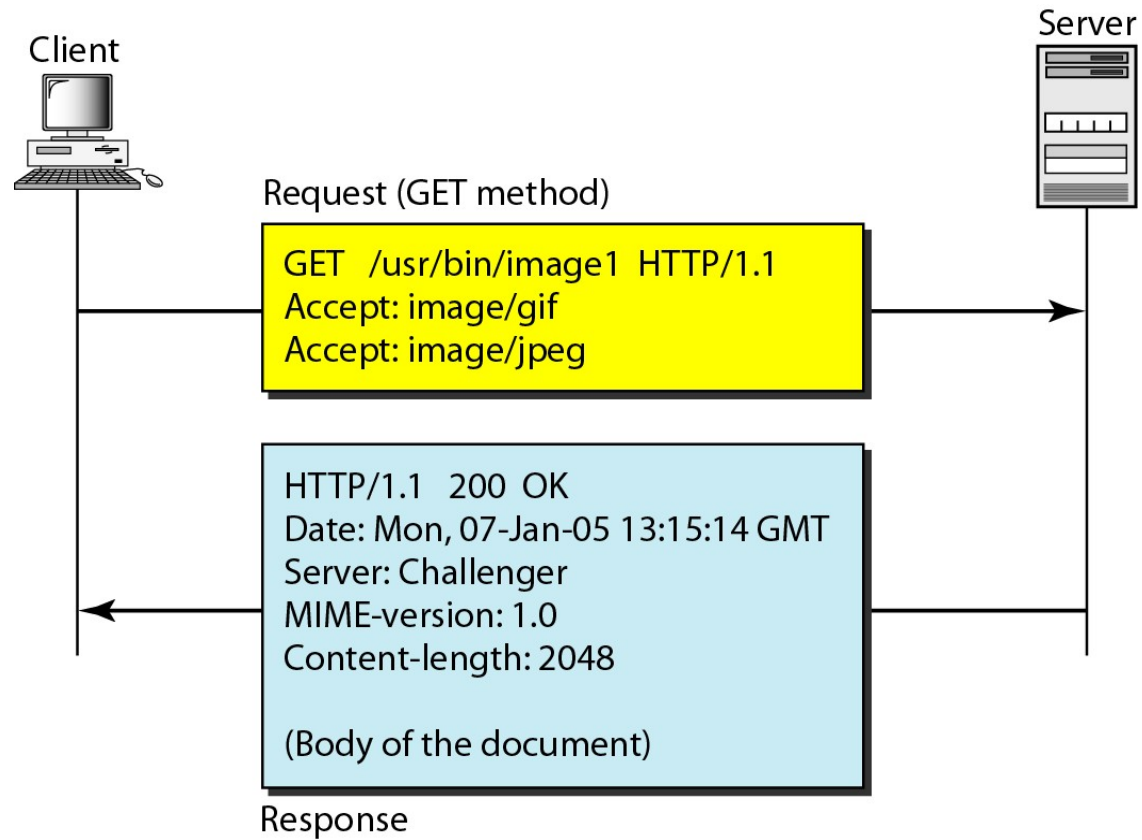
HTTP uses the services of TCP on well-known port 80



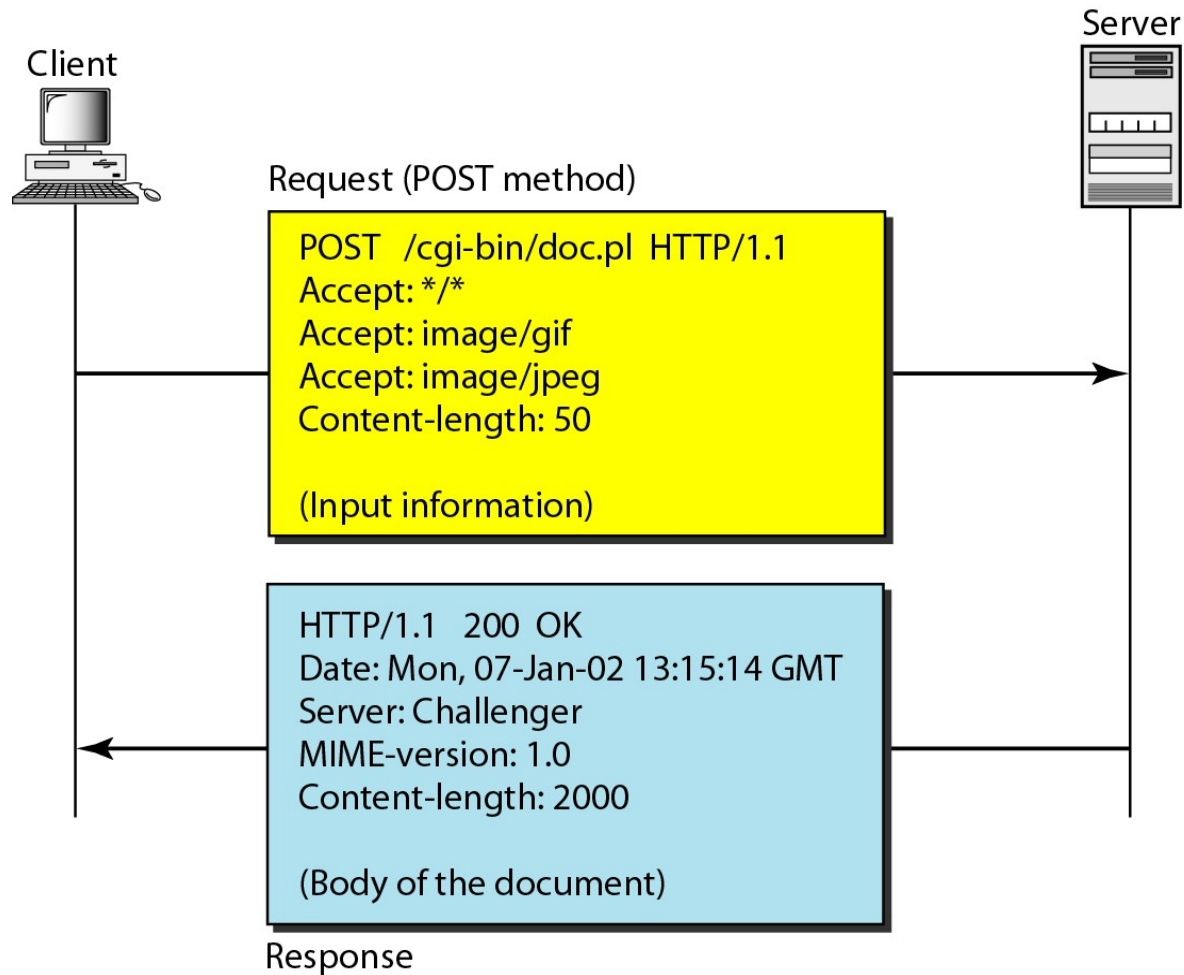
Request and Response Message



Example 1



Example 2



HTTP Methods

Method	Description
GET	Request to read a Web page
HEAD	Request to read a Web page's header
PUT	Request to store a Web page
POST	Append to a named resource (e.g., a Web page)
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Reserved for future use
OPTIONS	Query certain options

The built-in HTTP request methods.

HTTP Methods (2)

Code	Meaning	Examples
1xx	Information	100 = server agrees to handle client's request
2xx	Success	200 = request succeeded; 204 = no content present
3xx	Redirection	301 = page moved; 304 = cached page still valid
4xx	Client error	403 = forbidden page; 404 = page not found
5xx	Server error	500 = internal server error; 503 = try again later

The status code response groups.

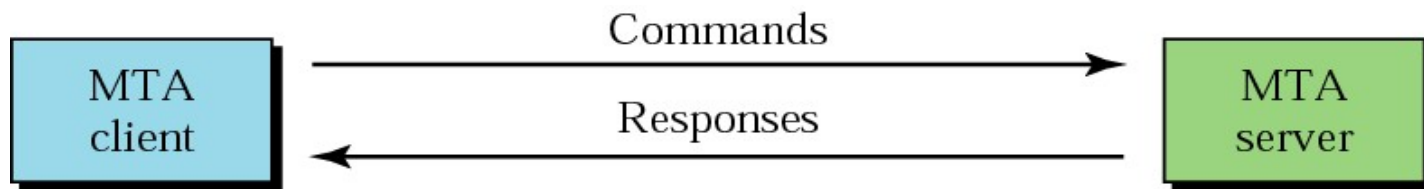
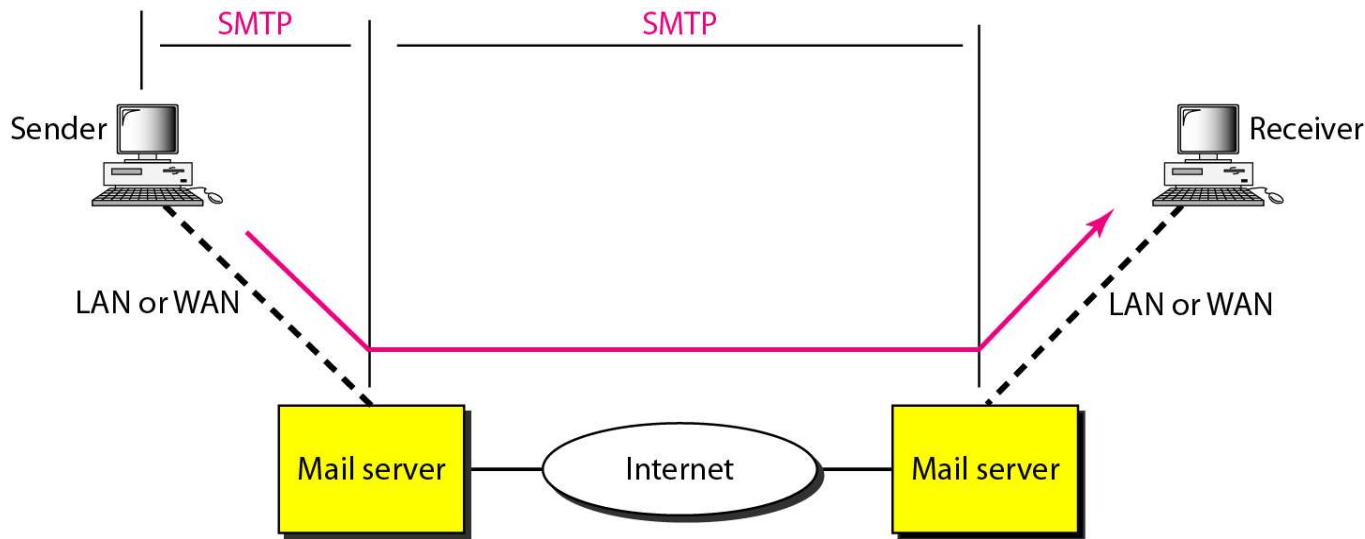
HTTP Message Headers

Header	Type	Contents
User-Agent	Request	Information about the browser and its platform
Accept	Request	The type of pages the client can handle
Accept-Charset	Request	The character sets that are acceptable to the client
Accept-Encoding	Request	The page encodings the client can handle
Accept-Language	Request	The natural languages the client can handle
Host	Request	The server's DNS name
Authorization	Request	A list of the client's credentials
Cookie	Request	Sends a previously set cookie back to the server
Date	Both	Date and time the message was sent
Upgrade	Both	The protocol the sender wants to switch to
Server	Response	Information about the server
Content-Encoding	Response	How the content is encoded (e.g., gzip)
Content-Language	Response	The natural language used in the page
Content-Length	Response	The page's length in bytes
Content-Type	Response	The page's MIME type
Last-Modified	Response	Time and date the page was last changed
Location	Response	A command to the client to send its request elsewhere
Accept-Ranges	Response	The server will accept byte range requests
Set-Cookie	Response	The server wants the client to save a cookie

Some HTTP message headers.

Mail Transfer Agent (MTA): SMTP

The actual mail transfer is done through MTA



Application Layer

- These SMTP commands specify the sender's and receiver's e-mail address, along with the message to be send.
- The exchange of commands between servers is carried out without intervention of any user.
- In case, message cannot be delivered, an error report is sent to the sender which makes SMTP a reliable protocol.

SMTP Commands

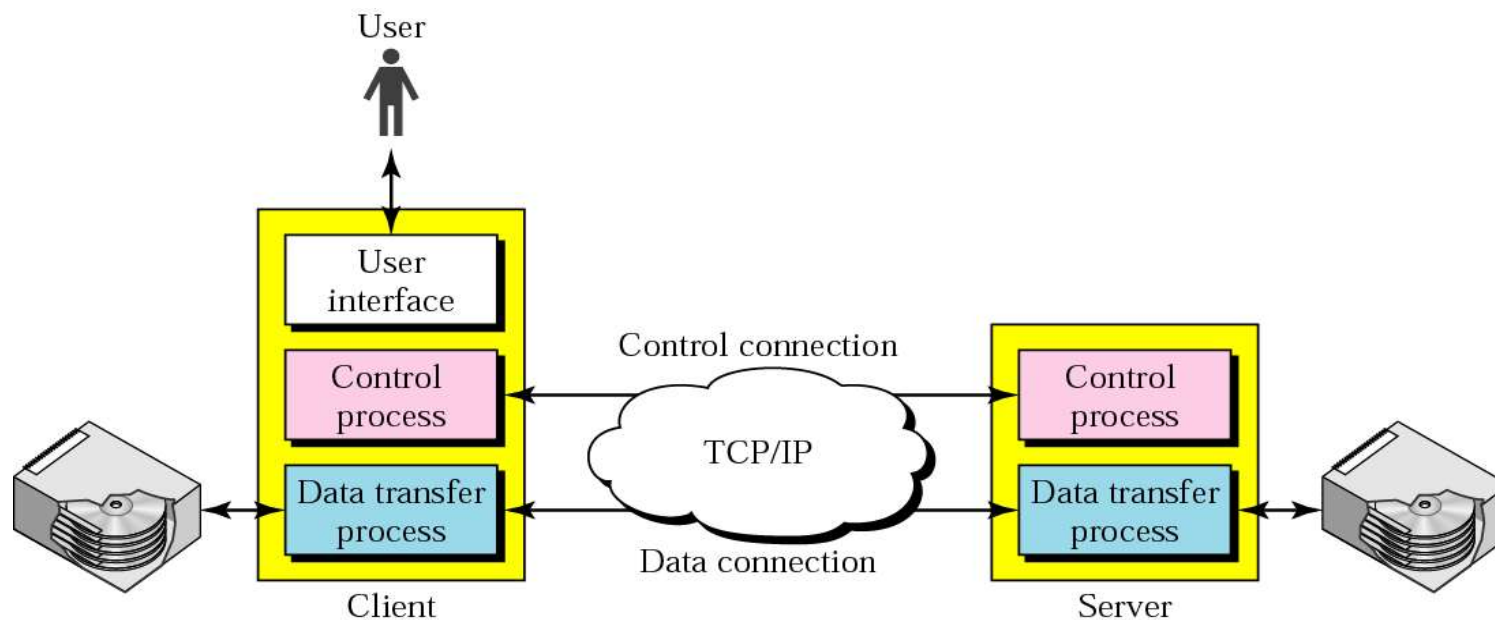
S.N o	Command	Description
1.	HELLO	This command initiates the SMTP conversation.
2.	EHELLO	This is an alternative command to initiate the conversation. ESMTP indicates that the sender server wants to use extended SMTP protocol.
3.	MAIL FROM	This indicates the sender's address.

Application Layer

S.N o	Command	Description
4.	RCPT TO	It identifies the recipient of the mail. In order to deliver similar message to multiple users this command can be repeated multiple times.
5.	SIZE	This command let the server know the size of attached message in bytes.
6.	DATA	The DATA command signifies that a stream of data will follow. Here stream of data refers to the body of the message.
7.	QUIT	This commands is used to terminate the SMTP connection.
8.	VERFY	This command is used by the receiving server in order to verify whether the given username is valid or not.
9	EXPN	It is same as VRFY, except it will list all the users name when it used with a distribution list.

File Transfer

FTP uses the services of TCP. It needs two TCP connections. The well-known port 21 is used for the control connection, and the well-known port 20 is used for the data connection.



Communication over Two Connections

